23CS55C – ARTIFICIAL INTELLIGENCE

CO3 - CODING ASSIGNMENT

Problem Identification (5Marks)	Algorithm Design (5Marks)	Completeness of Solution (5Marks)	Total (15 Marks)

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Batch:II

Assignment: Theorem Proving Using Z3 Solver

Question 40 – Autonomous Vehicle

Axiom 1: Every autonomous vehicle equipped with sensors can detect obstacles.
Axiom 2: Every vehicle that detects obstacles can plan a safe path.
Axiom 3: Every vehicle that plans a safe path can avoid obstacles.
Axiom 4: Vehicle1 is an autonomous vehicle equipped with sensors.
Theorem: Prove that Vehicle1 can avoid obstacles.

Meaning:

Domain → Object
The domain contains objects/individuals such as vehicles and autonomous vehicles.

Predicates → Autonomous(x), Sensor(x), Detect(x), Plan(x), Avoid(x)
• Autonomous(x): x is an autonomous vehicle.
• Sensor(x): x is equipped with sensors.
• Detect(x): x can detect obstacles.
• Plan(x): x can plan a safe path.
• Avoid(x): x can avoid obstacles.

Constant → Vehicle1
• Vehicle1: A particular individual (constant).

Axiom 1 → $\forall x$ ((Autonomous(x) $\wedge$ Sensor(x)) → Detect(x))
Axiom 1: For every individual, if that individual is an autonomous vehicle and is equipped with sensors, then it can detect obstacles.

Axiom 2 → $\forall x$ (Detect(x) → Plan(x))
Axiom 2: Every vehicle that detects obstacles can plan a safe path.

Axiom 3 → $\forall x$ (Plan(x) → Avoid(x))
Axiom 3: Every vehicle that plans a safe path can avoid obstacles.

Axiom 4 → Autonomous(Vehicle1) $\wedge$ Sensor(Vehicle1)
Axiom 4: Vehicle1 is an autonomous vehicle equipped with sensors.

Goal → Avoid(Vehicle1)
Goal: Prove that Vehicle1 can avoid obstacles.

Logical Reasoning:

From Axiom 4:
Autonomous(Vehicle1) $\wedge$ Sensor(Vehicle1)

From Axiom 1:
(Autonomous(Vehicle1) $\wedge$ Sensor(Vehicle1)) → Detect(Vehicle1)

Therefore, by Modus Ponens:
Detect(Vehicle1)

From Axiom 2:

Detect(Vehicle1) → Plan(Vehicle1)

Therefore, by Modus Ponens:
Plan(Vehicle1)

From Axiom 3:
Plan(Vehicle1) → Avoid(Vehicle1)

Therefore, by Modus Ponens:
Avoid(Vehicle1)

Therefore:
Vehicle1 can avoid obstacles.

Code:

```
from z3 import *


Object = DeclareSort('Object')


Autonomous = Function('Autonomous', Object, BoolSort())

EquippedWithSensors = Function('EquippedWithSensors', Object, BoolSort())

DetectsObstacles = Function('DetectsObstacles', Object, BoolSort())

PlansSafePath = Function('PlansSafePath', Object, BoolSort())

AvoidsObstacles = Function('AvoidsObstacles', Object, BoolSort())


vehicle1 = Const('Vehicle1', Object)

s = Solver()

x = Const('x', Object)


s.add(ForAll(x, Implies(And(Autonomous(x), EquippedWithSensors(x)),
DetectsObstacles(x))))

s.add(ForAll(x, Implies(DetectsObstacles(x), PlansSafePath(x))))

s.add(ForAll(x, Implies(PlansSafePath(x), AvoidsObstacles(x))))

s.add(Autonomous(vehicle1))

s.add(EquippedWithSensors(vehicle1))
```

```
goal = AvoidsObstacles(vehicle1)
s.add(Not(goal))

result = s.check()

if result == unsat:
    print("Theorem Proved: Vehicle1 can avoid obstacles.")
else:
    print("Counterexample found / Not provable:", s.model())
```

Output:

